

Xinfeng Xu

📍 Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA)

📍 NSF-Simons SkAI Institute (SkAI)

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Highlights

Research Expertise: Integrating astrophysics, data science, and statistics to study galaxy evolution, feedback effects, and the formation of the earliest galaxies and black holes.

Accomplishments: **15** first-author and **60+** co-author journal publications, **2** top-tier computer science conference papers (KDD, CIKM), **3300+** citations; **7** PI proposals (including **1 JWST** and **1 HST**) and **20+** Co-I proposals; granted **~\$0.6 million** as Science PI and **~\$5.5 million** as Science Co-I.

Educations

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| Ph.D. in Physics , <u>Virginia Tech</u> | <i>2014 – 2020</i> |
| ◦ Advisor: Prof. Nahum Arav | |
| ◦ Thesis: <i>How Do Quasars Impact Their Host Galaxies? Quasar Outflows Studies in Absorption and Emission</i> | |
| M.S. in Computer Science (double degree), <u>Virginia Tech</u> | <i>2016 - 2019</i> |
| ◦ Advisor: Prof. B. Aditya Prakash | |
| ◦ Thesis: <i>Modeling and Predicting Incidence using Deep Learning</i> | |

Appointments

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| Core Member | <i>2025 – present</i> |
| <i>NSF-Simons SkAI Institute, Evanston, IL; leading AI+astronomy research</i> | |
| CIERA Postdoctoral Fellow | <i>2023 – present</i> |
| <i>Northwestern University, Evanston, IL; focusing on astrophysics research</i> | |
| Postdoctoral Associate | <i>2020 – 2023</i> |
| <i>Johns Hopkins University, Baltimore, MD; focusing on space and astrophysics research</i> | |
| <i>Advisors: Prof. Tim Heckman (JHU), Dr. Alaina Henry (STScI)</i> | |

Principal Investigator Space-based Telescope Grants

Awarded:

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| James Webb Space Telescope (JWST) General Observing Program 5293, \$226,745 | <i>2025 – 2027</i> |
| <i>Galactic Winds in the Early Universe: observing outflows in emission and absorption in a typical $z \sim 6$ galaxy</i> | |
| Hubble Space Telescope (HST) Archival Research Program 17042, \$125,317 | <i>2023 – 2025</i> |
| <i>Are Galactic Outflows Seen in Absorption and Emission Lines Tracing the Same Gas?</i> | |

Recently Submitted:

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| JWST General Observing Program , requesting ~\$200k | <i>2025 Fall</i> |
| JWST General Observing Program , requesting ~\$500k | <i>2025 Fall</i> |

Principal Investigator Ground-based Telescope Grants

Ground-based telescopes typically do not award funding in addition to the observing time needed to conduct the proposed science, but the current “market rate” for purchasing time on these facilities is ~50–100k per night. Thus, my awarded PI time is equivalent to approximately ~250k in observational funding.

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| Keck Observatory, 2025B; with KCWI IFU Instrument | <i>1.0 night</i> |
| <i>Mapping the Dynamic Gaseous Environments in Lyman Continuum Leakers</i> | |

Keck Observatory, 2023B, 2024A, 2025A; with MOSFIRE Spectrograph <i>Local to Cosmic Noon: Are Galactic Outflows Seen in Absorption and Emission Lines Tracing the Same Gas?</i>	1.5 night
Gemini Observatory, 2022B; with NIFS IFU instrument <i>First-ever Mapping [OIII] Outflows in a Broad Absorption Line Quasar with Strong AGN Feedback</i>	0.5 night
Michigan-Dartmouth-MIT Observatory, 2021 <i>Unusual Balmer Decrements in Local Starburst Galaxies: Incorrect Calibration or New Astrophysics?</i>	2 nights

Co-Investigator Proposals and Grants

*Listing accepted open-access observatory proposals on which I served as a Co-I. These programs collectively represent approximately **\$5.5 million** in awarded federal funding, including about **\$330k that directly funded my research activities**. Depending on the project, I contributed to proposal preparation, data analysis, and/or scientific publications.*

HST Cycle 32, GO (11 orbits, PI: Arav, Nahum) <i>The largest BAL acceleration: testing disk wind models</i>	2025
HST Cycle 32, AR (PI: Arav, Nahum) <i>Measuring major contributors to AGN feedback: Quasar outflows in the HST/UV archive</i>	2025
HST Cycle 32, AR (PI: Henry, Alaina) <i>Mocking Galactic Winds: A new approach to constraining feedback by modeling the UV spectra of galaxies</i>	2025
JWST Cycle 4 (78.7 hrs; PI: Berg, Danielle) <i>The CLASSYIR Treasury: Unveiling the Cosmic Engines Powering Galaxies with JWST/MIRI</i>	2025
JWST Cycle 3 (22.65 hrs; PI: Vanzella, Eros) <i>Mapping Star Cluster Feedback in a Galaxy 500 Myr after the Big Bang</i>	2024
JWST Cycle 2 (23.99 hrs; PI: Mingozi, Matilde) <i>Tracing molecular gas in nearby metal-poor systems: the keys to unlocking star-formation in the early universe</i>	2023
JWST Cycle 2 (10.06 hrs; PI: Bradley, Larry) <i>Unveiling the Most Distant Lensed Arc at $z \sim 10$</i>	2023
JWST Cycle 2 (16.14 hrs; PI: Abdurro'uf) <i>Physical Properties of a Possible Galaxy Merger at $z=10.2$</i>	2023
JWST Cycle 1 (74.3 hrs; PI: Kassin, Susan) <i>A Pathfinder for JWST Spectroscopy: Deep High Spectral Resolution Maps of Galaxies over $1 < z < 6$</i>	2021
HST Cycle 31, AR Legacy (PI: Henry, Alaina) <i>Galactic Winds Unveiled: Leveraging Cloud Simulations with Radiative Transfer to Constrain Feedback</i>	2023
HST Cycle 31 (5 orbits; PI: Roy, Namrata) <i>Dissecting Red geyser winds: low luminosity AGNs with large scale outflows in the ionized phase</i>	2023
HST Cycle 31, AR (PI: Arav, Nahum) <i>Quasar outflows in the HST/UV archive: Measuring major contributors to AGN feedback</i>	2023
HST Cycle 30 (119 orbits; PI: Hayes, Matthew) <i>The Lyman-alpha and Continuum Origins Survey (LaCOS)</i>	2022
HST Cycle 30 (49 orbits; PI: Leclercq, Floriane) <i>Resolving Lyman Alpha emission in a complete sample of Lyman Continuum leakers and non-leakers</i>	2022
HST Cycle 30 (12 orbits; PI: Heckman, Timothy) <i>Are There Two Classes of Lyman-Leaky Galaxies?</i>	2022

HST Cycle 29, AR (PI: Carr, Cody) <i>Modeling the MgII-Lyman Alpha Relation as a Calibrator of the Lyman Continuum Escape Fraction</i>	2021
HST Cycle 29, AR (PI: Arav, Nahum) <i>Measuring the contribution of quasar outflows to AGN feedback</i>	2021
HST Cycle 29, AR (PI: Arav, Nahum) <i>A new paradigm for Seyfert outflows and their connection to AGN feedback</i>	2021
HST Cycle 29 (12 orbits; PI: James, Bethan) <i>[CII], a High-z Diagnostic Diamond in the Rough</i>	2021
HST Cycle 29 (14 orbits; PI: Hayes, Matthew) <i>The ionizing output of galaxies undergoing the most extreme feedback</i>	2021
HST Cycle 27, AR (PI: Arav, Nahum) <i>Are quasar outflows a major contributor to AGN feedback? HST/COS to the rescue</i>	2019
HST Cycle 24, AR (PI: Arav, Nahum) <i>The COS revolution of AGN outflow science</i>	2016
HST Cycle 24 (20 orbits; PI: Arav, Nahum) <i>Deciphering quasar outflows and measuring their contribution to AGN feedback</i>	2016
VLA/21B (160.17 hrs; PI: Borthakur, Sanchayeeta) <i>Characterizing Radio Continuum Emission from Low-z Lyman Continuum Leakers</i>	2021
Keck (2 nights; PI: Jaskot, Anne) <i>The Nebular Properties of Lyman Continuum Emitters: Deep Spectroscopy for the HST Low-Redshift Lyman Continuum Survey</i>	2020

Honors & Awards

CIERA Fellow , Northwestern University	2023
Kavli Fellow , Kavli Institute for the Physics and Math. of the Univ., Kashiwa, Japan	<i>declined</i> , 2023
STScI Director Research Fund , Space Telescope Science Institute	2022
Outstanding Master Thesis Award , Virginia Tech	2019
Graduate Student Scholarship , Virginia Tech	2017

Scientific Leadership Roles

The COS Legacy Archive Spectroscopic SurveY (CLASSY), project leader
CLASSY is an HST large treasury program (135 new + 177 archival orbits) with an international team of 45 experts to build a far-UV atlas of 45 star-forming galaxies & deliver high-level data products to the community. I led the studies of galactic outflow and feedback effects and published two first-author papers.

Low-z Lyman Continuum Survey (LzLCS), project leader
LzLCS is an HST large program (134 orbits) to statistically study the escape of ionizing Lyman continuum (LyC) radiation in 66 low-redshift star-forming galaxies. I led the studies of connecting MgII resonant emission lines to the escaped LyC and Ly α radiations and published one first-author paper.

Subaru Prime Focus Spectrograph (PFS), member
The PFS Galaxy Evolution Survey will observe 250,000+ distant galaxies over ~ 120 nights, the largest sample of such galaxies to date. I participate in the key aspects of the survey design, definition of the science case, SSP observations, data analyses (already started in Spring 2025!). I'm one of the project leads in the Galaxy Evolution group of PFS to study gas flows in star-forming galaxies and AGNs.

Habitable Worlds Observatory (HWO), science case development lead
HWO is NASA's next generation large infrared+optical+ultraviolet space telescope, recommended by the 2020 Decadal Survey. I led the science case development for "Spatially Resolving the Fundamental Elements of Reionization in Galaxies" to promote the UV IFU instruments on HWO.

Mentoring

Graduate Student:

- Tiffany Liou, Northwestern (M.S., current)

Undergraduate student:

- Yijun Song, Virginia Tech (2019 - 2020), now a graduate student at Purdue University
- Collier Sean, Virginia Tech (2018 - 2019), now assistant research professor (in Acoustics) at PSU
- Sean Heston, Virginia Tech (2018 - 2019), now a Ph.D. student at VT
- Jake Pighini, Emory and Henry College, VT REU student (2018), now a data scientist at Group W

High-school student:

- Caroline Baccus, through Northwestern/REACH program (current), becoming an undergraduate student in New York University in Winter 2025

Scientific & Community Service

Scientific Service

- SkAI Institute Astronomy + AI Joint Journal Club, organizer 2025
- Habitable World Observatory (HWO) Science Case Development, lead 2025
- CIERA Research Experience For Undergraduates (REU) selection committee, member 2024
- CIERA Journal Club, organizer 2024 - 2025
- James Webb Space Telescope (JWST) Time Allocation Committee, Panel Support Scientist 2023
- Hubble Space Telescope (HST), Time Allocation Committee, external panelist 2022, 2023
- Gemini Telescope, Fast Turnaround Time Allocation Committee, external panelist 2021, 2022
- UV Galaxies Conference at Iceland, Organizing committee member and session chair 2023
- Large Volume Spectroscopic Analysis Conference at JWST Era, Organizing committee member 2022
- Johns Hopkins University/Space Telescope Science Institute Astrocoffee, organizer 2021 - 2023

Community Service

- Peer reviewer for The Astrophysical Journal (ApJ), Monthly Notices of the Royal Astronomical Society (MNRAS), and Astronomy & Astrophysics 2020 - now
- CIERA Equity, Diversity, and Inclusion Communication and Visibility Action Team, member 2024

Teaching

Lecture: including course design, material preparations, class involvements, and assessment.

- “*Introduction to Modern Astronomical Telescopes*”, undergraduate class (1 credit) at Johns Hopkins 2023

Lectures: including material preparations and class involvements

- “*Advanced Topics in Astrophysics: Galaxy Formation and Evolution*”, graduate class at Northwestern 2024
- “*Introduction to Cosmology*”, CIERA REACH program class at Northwestern 2024

Scientific Talks

Invited Talks at:

- University of Texas at Dallas, seminar Apr 2025
- The SkAI Institute, Joint Astronomy + AI Journal Club Mar 2025
- University of California Los Angeles, seminar Dec 2024
- California Institute of Technology, seminar Dec 2024
- Carnegie Observatories, seminar Dec 2024
- Massachusetts Institute of Technology, seminar Feb 2024
- University of Science and Technology of China, seminar Jul 2023

◦ UV Galaxies Conference at Iceland	Jul 2023
◦ Princeton University, seminar	Mar 2022
◦ Johns Hopkins University/Space Telescope Science Institute, astrocoffee	Sep 2021
◦ Johns Hopkins University/Space Telescope Science Institute, joint seminar	Feb 2021
◦ Space Telescope Science Institute, Joint Seminar	Jun 2020
◦ York University, seminar	Sep 2019
◦ Virginia Tech, seminar	Apr 2018

Contributed Conference Talks or Posters at:

◦ “The Formation of the Earliest Cosmic Structures” at Cosmic Frontier Center (CFC)	May 2025
◦ “Winds throughout the Universe” at Joint Space-Science Institute (JSI)	Oct 2023
◦ “241st American Astronomical Society (AAS) Meeting” at Seattle	Jan 2023
◦ “LymanRAS: The production and escape of Lyman photons through time and space”, virtual	Jan 2022
◦ “SAZERAC: Early Galaxy Formation Near and Far: Preparing for a Journey with JWST”, virtual	Dec 2021
◦ “Galactic and AGN Wind Workshop” at Baltimore	Aug 2021
◦ “25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining”	Aug 2019
◦ “233rd American Astronomical Society (AAS) Meeting” at Seattle	Jan 2019
◦ “231st American Astronomical Society (AAS) Meeting” at Washington D.C.	Jan 2018
◦ “Conference on Information and Knowledge Management (CIKM)”	Nov 2017
◦ “AGN Winds Conference on the Georgia Coast”	Jun 2017

Selected Press Releases

All based on research from my journal publications.

@ESA, Webb captures star clusters in Cosmic Gems arc 	2024
@Astronomy.com, New CLASSY atlas provides clues about galaxy evolution 	2022
@NASA, Quasar Tsunamis Rip Across Galaxies 	2020
@VT news, Discover quasar tsunamis capable of preventing stars from forming 	2020

Publications as first authors or led by my students

17. Liou, Tiffnay; **Xu, Xinfeng**; Strom; Allison; et al.
Galactic Outflow Properties at Cosmic Noon: Faster, Stronger but Consistent Feedback Effects
Submitted to The Astrophysical Journal
16. Baccus, Caroline J.; **Xu, Xinfeng**; et al.
A Comprehensive JWST/NIRSpec Census of Broad-Line Active Galactic Nuclei: Faint, Tiny, but Highly Accreting Sources in the Remote Universe
Submitted to The Astrophysical Journal
15. **Xu, Xinfeng**; Fielding, Drummond; Heckman, Timothy; Bryan, Greg; et al.
[Resolving the Unresolved Galactic Winds in Multi-phase Models. I. Methodology and Application](#) 
The Astrophysical Journal, 2026ApJ..1003..155X
14. **Xu, Xinfeng**; McCandliss, Stephan; Strom, Allison; Hsiao-Wen, Chen, Choi, Yumi; et al.
[Spatially Resolving the Fundamental Elements of Reionization in Galaxies](#) 
Astronomical Society of the Pacific (ASP) Conference Proceedings
13. **Xu, Xinfeng**; Henry, Alaina; Heckman, Timothy; Carr, Cody; et al.
[Shining a Light on the Connections between Galactic Outflows Seen in Absorption and Emission Lines](#) 
The Astrophysical Journal, 2025ApJ...984...94X
12. **Xu, Xinfeng**; Heckman, Tim; Yoshida, Michitoshi; Ohyama, Youichi; Henry, Alaina
[What are the Radial Distributions of Density, Outflow Rates, and Cloud Structures in the M 82 Wind?](#) 
The Astrophysical Journal, 2023ApJ...956..142X

11. **Xu, Xinfeng**; Henry, Alaina; Heckman, Tim; Marques-Chaves, Rui; et al.
[The Low-Redshift Lyman Continuum Survey: Optically Thin and Thick Mg II Lines as Probes of Lyman Continuum Escape](#) 
The Astrophysical Journal, 2023ApJ...943...94X
10. **Xu, Xinfeng**; Heckman, Tim; Henry, Alaina; Berg, Danielle A.; et al.
[CLASSY VI: The Density, Structure and Size of Absorption-Line Outflows in Starburst Galaxies](#) 
The Astrophysical Journal, 2023ApJ...948...28X
9. **Xu, Xinfeng**; Henry, Alaina; Heckman, Tim; Chisholm, John et al.
[Tracing Ly \$\alpha\$ and Lyman Continuum Escape in Galaxies by Mg II Emission](#) 
The Astrophysical Journal, 2022ApJ...933...202. (2022b)
8. **Xu, Xinfeng**; Heckman, Tim; Henry, Alaina; Berg, Danielle A.; Chisholm, John; James, Bethan L.; Martin, Crystal L.; Stark, Daniel P. and the CLASSY Team;
[The COS Legacy Archive Spectroscopy SurVeY \(CLASSY\) III: The Properties of Starburst-Driven Warm Ionized Outflows](#) 
The Astrophysical Journal, 2022ApJ...933..222X (2022a)
7. **Xu, Xinfeng**; Arav, Nahum; Miller, Timothy; Korista, Kirk T.; Benn, Chris;
[Physical Conditions of Iron-peak Low-ionization Lines in the FeLoBAL Quasar Q0059-2735](#) 
Monthly Notices of the Royal Astronomical Society, 2021MNRAS.506.2725X.
6. **Xu, Xinfeng**; Zakamska, Nadia L.; Arav, Nahum; Miller, Timothy; Benn, Chris;
[Evidence that Emission and Absorption Outflows in Quasars Are Related](#) 
Monthly Notices of the Royal Astronomical Society, 2020MNRAS.495..305X.
5. **Xu, Xinfeng**; Arav, Nahum; Miller, Timothy; Kriss G. A.; Plesha, R.;
[HST/COS observations of quasar outflows in the 500 – 1050Å rest-frame, II: The Most Energetic Quasar Outflow](#) 
The Astrophysical Journal Supplement, 2020ApJS..247...38X.
4. **Xu, Xinfeng**; Arav, Nahum; Miller, Timothy; Kriss G. A.; Plesha, R.;
[HST/COS observations of quasar outflows in the 500 – 1050Å rest-frame, IV: The Largest Broad Absorption Line Acceleration](#) 
The Astrophysical Journal Supplement, 2020ApJS..247...40X
3. **Xu, Xinfeng**; Arav, Nahum; Miller, Timothy; Kriss G. A.; Plesha, R.;
[HST/COS observations of quasar outflows in the 500 – 1050Å rest-frame, VI: Wide, Energetic Outflows in SDSS J0755+2306](#) 
The Astrophysical Journal Supplement, 2020ApJS..247...42X
2. **Xu, Xinfeng**; Arav, Nahum; Miller, Timothy; Benn, Chris
[VLT/X-Shooter Survey of BAL Quasars: Large Distance Scale and AGN Feedback](#) 
The Astrophysical Journal, 2019ApJ...876..105X.
1. **Xu, Xinfeng**; Arav, Nahum; Miller, Timothy; Benn, Chris
[A Mini-BAL Outflow at 900 pc from the Central Source: VLT/X-shooter Observations](#) 
The Astrophysical Journal, 2018ApJ...858...39X

Other Co-authored Publications

53. Abdurro'uf; Coe, Dan; Resseguier, Tom; Murphy, Calla; Xu, Xinfeng; et al.
[Spatially Resolved Physical Properties of Young Star Clusters and Star-forming Clumps in the Brightest z \$\approx\$ 6 Galaxy, the Strongly Lensed Cosmic Spear at z=6.2](#) 
Submitted to The Astrophysical Journal, 2025arXiv251208054A
52. Komarova, Lena; Oey, Sally; Marques-Chaves, Rui; Amorín, Ricardo; Henry, Alaina; et al.
[Power-law Emission-line Wings and Radiation-driven Superwinds in Local Lyman Continuum Emitters](#) 
The Astrophysical Journal, 2025ApJ...994...192

51. Arellano-Córdova, K. Z.; Berg, D. A.; Mingozi, M.; James, B. L.; Vincenzo, F.; et al.
[CLASSY XII: Nitrogen Enrichment Shaped by Gas Density and Feedback](#) 
The Monthly Notices of the Royal Astronomical Society, 2025MNRAS.tmp.1623A
50. Roy, Namrata; Heckman, Timothy; Henry, Alaina; Chisholm, John; Flury, Sophia; et al.
[Lyman Continuum Leakage from Massive Leaky Starbursts: A Different Class of Emitters?](#) 
The Astrophysical Journal, 2025ApJ...992...91R
49. Le Reste, Alexandra; Jaskot, Anne E.; Brazie, Jordanne; Scarlata, Claudia; Flury, Sophia R.; et al.
[The Ly \$\alpha\$ and Continuum Origins Survey. III. Investigating the link between galaxy morphology, merger properties and LyC escape](#) 
submitted to The Astrophysical Journal, arXiv:2509.06922
48. Saldana-Lopez, A.; Hayes, M. J.; Le Reste, A.; Scarlata, C.; Melinder, J.; Henry, A.; et al.
[The Lyman-alpha and Continuum Origins Survey II: the connection between the escape of ionizing radiation and Lyman-alpha halos in star-forming galaxies](#) 
submitted to The Astrophysical Journal, arXiv:2504.07074
47. Le Reste, Alexandra; Scarlata, Claudia; Hayes, Matthew J.; Melinder, Jens; Saldana-Lopez, Alberto; et al.
[The Ly \$\alpha\$ and Continuum Origins Survey. I. Survey Description and Ly \$\alpha\$ Imaging](#) 
submitted to The Astrophysical Journal, 2025ApJS..280...27L
46. Messa, M.; Vanzella, E.; Loiacono, F.; Adamo, A.; Oguri, M.; Sharon, K.; Bradley, L. D; et al.
[JWST Spectroscopic Confirmation of the Cosmic Gems Arc at \$z=9.625\$ – Insights into the small scale structure of a post-burst system](#) 
submitted to The Astrophysical Journal, arXiv:2507.18705
45. Vanzella, E.; Messa, M.; Adamo, A.; Loiacono, F.; Oguri, M.; Sharon, K.; Bradley, L. D.; et al.
[The \$z = 9.625\$ Cosmic Gems Galaxy was a "Compact Blue Monster" Propelled by Massive Star Clusters](#) 
submitted to The Astrophysical Journal, arXiv:2507.18699
44. Komarova, Lena; Oey, Sally; Marques-Chaves, Rui; Amorín, Ricardo; Henry, Alaina; Schaerer, Daniel; et al.
[Power-law Emission-line Wings and Radiation-Driven Superwinds in Local Lyman Continuum Emitters](#) 
submitted to The Astrophysical Journal, arXiv:2506.19623
43. Flury, Sophia R.; Jaskot, Anne E.; Saldana-Lopez, Alberto; Oey, M. S.; Chisholm, John; et al.
[The Low-redshift Lyman Continuum Survey: The Roles of Stellar Feedback and Interstellar Medium Geometry in LyC Escape](#) 
The Astrophysical Journal, 2025ApJ...985..128F
42. Carr, Cody A.; Cen, Renyue; Scarlata, Claudia; **Xu, Xinfeng**; et al.
[The Effect of Radiation and Supernovae Feedback on LyC Escape in Local Star-forming Galaxies](#) 
The Astrophysical Journal, 2025ApJ...982..137C
41. Peng, Zixuan; Martin, Crystal L.; Chen, Zirui; Fielding, Drummond B.; **Xu, Xinfeng**; et al.
[Physical Origins of Outflowing Cold Clouds in Local Star-forming Dwarf Galaxies](#) 
The Astrophysical Journal, 2025ApJ...981..171P
40. Jennings, R. Michael; Henry, Alaina; Mauerhofer, Valentin; Heckman, Timothy; et al.
[A Simulated Galaxy Laboratory: Exploring the Observational Effects on UV Spectral Absorption Line Measurements](#) 
The Astrophysical Journal, 2025ApJ...979...64J
39. Parker, Kaelee S.; Berg, Danielle A.; Gazagnes, Simon; et al.
[CLASSY. XI. Tracing Neutral Gas Properties Using UV Absorption Lines and 21 cm Observations](#) 
The Astrophysical Journal, 2024ApJ...977..104P
38. Li, Zhihui; Gronke, Max; Heckman, Timothy; **Xu, Xinfeng**; et al.
[Synergistic Radiative Transfer Modeling of Mg II and Ly \$\alpha\$ Emission in Multiphase, Clumpy Galactic Environments: Application to Low-Redshift Lyman Continuum Leakers](#) 
The Astrophysical Journal, 2025ApJ...991...49L

37. Roy, Namrata; Heckman, Timothy; Henry, Alaina; Chisholm, John; et al.
[Lyman Continuum leakage from massive leaky starbursts: A different class of emitters?](#) 
Submitted to The Astrophysical Journal, 2024arXiv241013254R
36. Bradley, Larry D.; Adamo, Angela; Vanzella, Eros; Sharon, Keren; Brammer, Gabriel; Coe, Dan; et al.
[Unveiling the Cosmic Gems Arc at \$z \sim 10.2\$ with JWST](#) 
The Astrophysical Journal, 2025ApJ...991...32B
35. Hsiao, Tiger Yu-Yang; Álvarez-Márquez, Javier; Coe, Dan; et al.
[JWST MIRI Detections of H \$\alpha\$ and \[O III\] and a Direct Metallicity Measurement of the \$z = 10.17\$ Lensed Galaxy MACS0647-JD](#) 
The Astrophysical Journal, 2024ApJ...973...81H
34. Flury, Sophia R.; Jaskot, Anne E.; Saldana-Lopez, Alberto; et al.
[The Low-Redshift Lyman Continuum Survey: The Roles of Stellar Feedback and ISM Geometry in LyC Escape](#) 
Accepted to The Astrophysical Journal, 2024arXiv240912118F
33. Huberty, M.; Carr, C.; Scarlata, C.; Heckman, T.; Henry, A.; Xu, X.; et al.
[CLASSY X: Highlighting Differences Between Partial Covering and Semi-Analytic Modeling in the Estimate of Galactic Outflow Properties](#) 
The Astrophysical Journal, 2024ApJ...975...58H
32. Jaskot, Anne E.; Silveyra, Anneliese C.; Plantinga, Anna; et al.
[Multivariate Predictors of LyC Escape II: Predicting LyC Escape Fractions for High-Redshift Galaxies](#) 
The Astrophysical Journal, 2024ApJ...973..111J
31. Jaskot, Anne E.; Silveyra, Anneliese C.; Plantinga, Anna; et al.
[Multivariate Predictors of Lyman Continuum Escape. I. A Survival Analysis of the Low-redshift Lyman Continuum Survey](#) 
The Astrophysical Journal, 2024ApJ...972...92J
30. Leclercq, Floriane; Chisholm, John; King, Wichahpi; et al.
[Linking Mg II and \[O II\] spatial distribution to ionizing photon escape in confirmed LyC leakers and non-leakers](#) 
Astronomy & Astrophysics, 2024A&A...687A..73L
29. Adamo, Angela; Bradley, Larry D.; Vanzella, Eros; et al.
[Bound star clusters observed in a lensed galaxy 460 Myr after the Big Bang](#) 
Nature, June 2024
28. Hu, Weida; Martin, Crystal L.; Gronke, Max; Gazagnes, Simon et al.
[CLASSY VII Ly \$\alpha\$ Profiles: The Structure and Kinematics of Neutral Gas and Implications for LyC Escape in Reionization-era Analogs](#) 
The Astrophysical Journal, 2023ApJ...956...39H
27. Gazagnes, Simon; Mauerhofer, Valentin; Berg, Danielle A. et al.
[Interpreting the Si II and C II Line Spectra from the COS Legacy Archive Spectroscopic Survey Using a Virtual Galaxy from a High-resolution Radiation-hydrodynamic Simulation](#) 
The Astrophysical Journal, 2023ApJ...952..164G
26. Mingozi, Matilde; James, Bethan L.; Berg, Danielle A.; Arellano-Córdova, Karla Z.; et al.
[CLASSY VIII: Exploring the Source of Ionization with UV ISM diagnostics in local High- \$z\$ Analogs](#) 
The Astrophysical Journal, 2024ApJ...962...95M
25. Hsiao, Tiger Yu-Yang; Abdurro'uf; Coe, Dan; Larson, Rebecca L.; Jung, Intae; Mingozi, Matilde; et al.
[JWST NIRSpec spectroscopy of the triply-lensed \$z = 10.17\$ galaxy MACS0647-JD](#) 
The Astrophysical Journal, 2024ApJ...973....8H

24. Fudamoto, Yoshinobu; Inoue, Akio K.; Coe, Dan; Welch, Brian; Acebron, Ana; Ricotti, Massimo; Mandelker, Nir; Windhorst, Rogier A.; **Xu, Xinfeng**; et al.
[The Extended \[CII\] under Construction? Observation of the brightest high-z lensed star-forming galaxy at \$z = 6.2\$](#) 
The Astrophysical Journal, 2024ApJ...961...71F
23. Abdurro'uf; Coe, Dan; Jung, Intae; Ferguson, Henry; Brammer, Gabriel; et al.
[Spatially Resolved Stellar Populations of \$0.3 < z < 6.0\$ Galaxies in WHL 0137-08 and MACS 0647+70 Clusters as Revealed by JWST: How Do Galaxies Grow and Quench over Cosmic Time?](#) 
The Astrophysical Journal, 2023ApJ...945..117A
22. Mingozi, Matilde; James, Bethan L.; Arellano-Córdova, Karla Z.; Berg, Danielle A.; et al.
[CLASSY IV: Exploring UV diagnostics of the interstellar medium in local high- \$z\$ analogs at the dawn of the JWST era](#) 
The Astrophysical Journal, 2022ApJ...939..110M
21. Arellano-Córdova, Karla Z.; Mingozi, Matilde; Berg, Danielle A.; et al.
[CLASSY V: The Impact of Aperture Effects on the Inferred Nebular Properties of Local Star-forming Galaxies](#) 
The Astrophysical Journal, 2022ApJ...935...74A
20. Chisholm, J.; Saldana-Lopez, A.; Flury, S.; Schaerer, D.; Jaskot, A. et al.
[The Far-Ultraviolet Continuum Slope as a Lyman Continuum Escape Estimator at High-redshift](#) 
Monthly Notices of the Royal Astronomical Society, 2022MNRAS.517.5104C
19. Marques-Chaves, R.; Schaerer, D.; Amorín, R. O.; Atek, H.; Borthakur, S.; et al.
[No correlation of the Lyman continuum escape fraction with spectral hardness](#) 
Astronomy & Astrophysics, 2022A&A...663L...1M
18. James, Bethan L.; Berg, Danielle A.; King, Teagan; and the CLASSY Team
[CLASSY-II: A technical Overview of the COS Legacy Archive Spectroscopic Survey](#) 
The Astrophysical Journal Supplement, 2022ApJS..262...37J
17. Berg, Danielle A.; James, Bethan L.; King, Teagan; McDonald, Meaghan; Chisholm, John; Heckman, Timothy; Martin, Crystal L.; Stark, Dan P. and the CLASSY Team
[The COS Legacy Archive Spectroscopy Survey \(CLASSY\) Treasury Atlas](#) 
The Astrophysical Journal Supplement, 2022ApJS..261...31B
16. Saldana-Lopez, Alberto; Schaerer, Daniel; Chisholm, John; Flury, Sophia R.; Jaskot, Anne E. et al.
[The Low-Redshift Lyman Continuum Survey: Unveiling the ISM properties of low- \$z\$ Lyman continuum emitters](#) 
Astronomy & Astrophysics, 2022A&A...663A..59S
15. Flury, Sophia R.; Jaskot, Anne E.; Ferguson, Harry C.; Worseck, Gábor; et al.
[The Low-redshift Lyman Continuum Survey. II. New Insights into LyC Diagnostics](#) 
The Astrophysical Journal, 2022ApJ...930..126F
14. Flury, Sophia R.; Jaskot, Anne E.; Ferguson, Harry; Worseck, Gabor; Makan, Karill; et al.
[The Low-redshift Lyman Continuum Survey. I. New, Diverse Local Lyman Continuum Emitters](#) 
The Astrophysical Journal, 2022ApJS..260....1F
13. Wang, Bingjie; Heckman, Timothy M.; Ricardo, Amorin; Sanchayeeta, Borthakur; Chisholm, John et al.
[The Low-redshift Lyman-continuum Survey: \[S II\] Deficiency and the Leakage of Ionizing Radiation](#) 
The Astrophysical Journal, 2021ApJ...916....3W.
12. Miller, Timothy; Arav, Nahum; **Xu, Xinfeng**; Kriss G. A.;
[The contribution of quasar absorption outflows to AGN feedback](#) 
Monthly Notices of the Royal Astronomical Society, 2020MNRAS.499.1522M
11. Zeilig-Hess, Meir; Levinson, Amir; **Xu, Xinfeng**; Arav, Nahum;
[BALQSO Spectra Explained by Shock Disruption of Galactic Clouds](#) 
Monthly Notices of the Royal Astronomical Society, 2020MNRAS.491.4325Z.

10. Miller, Timothy; Arav, Nahum; **Xu, Xinfeng**; Kriss G. A.; Plesha, R.;
[HST/COS Observations of Quasar Outflows in the Extreme UV, III: Four Similar and Energetic Outflows in 2MASS J1051+1247 Likely Contributing to AGN Feedback](#) 
The Astrophysical Journal Supplement, 2020ApJS..247...39M
9. Miller, Timothy; Arav, Nahum; **Xu, Xinfeng**; Kriss G. A.; Plesha, R.;
[HST/COS Observations of Quasar Outflows in the Extreme UV, V: Two Outflows in PKS J0352-0711: Distances, Energetics, and AGN Feedback](#) 
The Astrophysical Journal Supplement, 2020ApJS..247...41M
8. Miller, Timothy; Arav, Nahum; **Xu, Xinfeng**; Kriss G. A.; Plesha, R.;
[HST/COS Observations of Quasar Outflows in the 500-1050Å Rest Frame. VII. Distances and Energetics for 11 Outflows in Five Quasars](#) 
The Astrophysical Journal Supplement, 2020ApJS..249...15M
7. Arav, Nahum; **Xu, Xinfeng**; Kriss G. A. et al. (HST/COS collaboration, 21 co-authors)
[Multi-wavelength campaign on NGC 7469, V. Analysis of the HST/COS observations: Super solar metallicity, distance, and trough variation models](#) 
Astronomy & Astrophysics, 2020A&A...633A..61A
6. Arav, Nahum; **Xu, Xinfeng**; Miller, Timothy; Kriss G. A.; Plesha, R.;
[HST/COS observations of quasar outflows in the 500 – 1050Å rest-frame, I: The Most Energetic Quasar Outflows In The Universe And Other Discoveries](#) 
The Astrophysical Journal Supplement, 2020ApJS..247...37A
5. Kriss G. A. et al. (HST/COS, XMM-Newton and NuSTAR collaborations, 27 co-authors)
[HST/COS observations of the newly discovered obscuring outflow in NGC 3783](#) 
Astronomy & Astrophysics, 2019A&A...621A..12k
4. Adhikari, Bijaya; **Xu, Xinfeng**; Ramakrishnan, Naren; Prakash, B. Aditya;
[EpiDeep: Exploiting Embeddings for Epidemic Forecasting](#) 
Participated in U.S. national Centers for Disease Control and Prevention (CDC) flu challenge, 2017 – 2018
Published in Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, Pages 577-586, 2019
3. Chen, Liangzhe; **Xu, Xinfeng**; Lee, Sangkeun; Duan, Sisi; Tarditi, Alfonso G.; Chinthavali, Supriya; Prakash, B. Aditya;
[HotSpots: Failure Cascades on Heterogeneous Critical Infrastructure Networks](#) 
Collaborated with Oak Ridge National Laboratory (ORNL), US
Published in Proceedings of the Conference on Information and Knowledge Management (CIKM), Pages 1599-1607, 2017
2. Miller, Timothy R.; Arav, Nahum; **Xu, Xinfeng**; Kriss, Gerard A.; Plesha, Rachel J.; Benn, Chris; Liu, Guilin
[Distance, Energy, and Variability of Quasar Outflows: Two HST/COS Epochs of LBQS 1206+1052](#) 
The Astrophysical Journal, 2018ApJ...865...90M
1. Arav, Nahum; Liu, Guilin; **Xu, Xinfeng**; Stidham, James; Benn, Chris; Chamberlain, Carter
[Evidence that 50% of BALQSO Outflows Are Situated at Least 100 pc from the Central Source](#) 
The Astrophysical Journal, 2018ApJ...857...60